FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 17-11-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. $B_{net} = \frac{\mu_0 i}{4\pi R}\left(\frac{\sqrt{3}}{2}\right) - \frac{\mu_0 i}{4\pi R\sqrt{3}}\left(\frac{1}{2}\right)$

$= \frac{\mu_0 i}{4\sqrt{3}\pi R}$

2. D

Sol. Total charge (finally) = 0

So charge flown to the earth = $\sum_{i=1}^{N} Q_i$

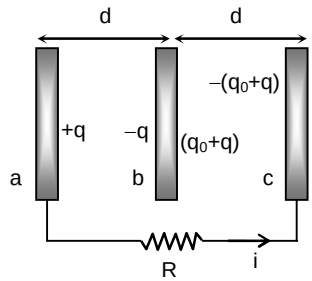
3. B

Sol. $\frac{q}{C} + \frac{q_0 + q}{C} = iR$ (where $C = \frac{\varepsilon_0 A}{d}$)

$\Rightarrow i = \frac{q_0 + 2q}{RC} \quad \Rightarrow -\frac{dq}{dt} = \frac{q_0 + 2q}{RC}$

$\Rightarrow -\int_0^q \frac{dq}{q_0 + 2q} = \int_0^t \frac{dt}{RC}$

$i = \frac{q_0 d}{R\varepsilon_0 A} e^{\frac{-2dt}{R\varepsilon_0 A}}$



4. C

Sol. $(P)\left(\frac{4}{3}\pi r^3\right) = \sigma\left(4\pi r^2\right)T^4$

So $T = cr^{1/4}$ (where c is a constant)

5. B

Sol. All the spheres are at same potential so, $V_1 = V_2 = V_3$ and $Q_1 : Q_2 : Q_3 = R_1 : R_2 : R_3$ and $E_1 : E_2 : E_3 = \frac{1}{R_1} : \frac{1}{R_2} : \frac{1}{R_3}$.

6. B

Sol. $\frac{dQ}{dt} = kA\frac{dT}{dx}$

7. B

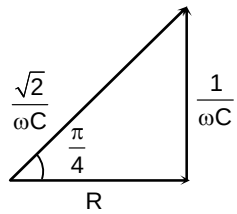
Sol. $W_{elect} = U_i - U_f$

$$U_i = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1^2}{2a} + \frac{q_2^2}{2b} + \frac{q_3^2}{2c} + \frac{q_1q_3}{c} + \frac{q_1q_2}{b} + \frac{q_2q_3}{c}\right)$$

$$U_f = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1^2}{2a} + \frac{q_2^2}{2b'} + \frac{q_3^2}{2c} + \frac{q_1q_3}{c} + \frac{q_1q_2}{b'} + \frac{q_2q_3}{c}\right)$$

8. B, C

Sol. From the Phasor diagram



9. A, D

Sol. Capacitors are connected in series, so charges on the capacitors remain same. During removal of the slab anticlockwise current flows through the circuit, so battery is being charged.

10. B

Sol. $0 = dU + PdV = d(a + bPV) + PdV$

After solving

$PV^{\frac{b+1}{b}} = \text{constant}$

11. A, B, C, D

Sol. When only S_2 is open

$\frac{2}{3} = \left(\frac{4}{120}\right)\ell \Rightarrow \ell = 20$ cm

When only S_1 is open

$2 = \left(\frac{4}{120}\right)\ell \Rightarrow \ell = 60$ cm

When all three are closed

$2 = \frac{4}{120} \times \ell \Rightarrow \ell = 60$ cm

When only S_1 is closed

$2 = \frac{4}{120} \times \ell \Rightarrow \ell = 60$ cm

12. B, D

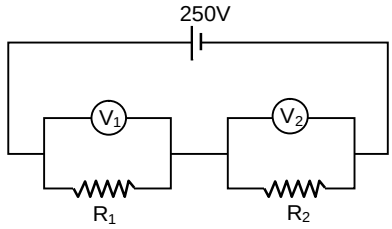
Sol. Every body radiates and absorbs radiation simultaneously at any non zero temperature T.

13. B, D

Sol. When switch S is open

$$V_1 = \frac{250}{10000}(6000) = 150\,V$$

$$V_2 = \frac{250}{10000}(4000) = 100\,V$$



When switch S is closed circuit reduces to shown in the figure.

So both the voltmeters show readings $V_1 = V_2 = \frac{250}{2} = 125\,V$

14. A, B

Sol. Distance of the target x = (n) $\frac{2\pi m}{qB} v\cos\theta$, where n is an integer

$$\frac{nv}{B} = \frac{v_0}{B_0}$$

15. A

16. B

Sol. (for Q.15-16)

Emf induced between the plates = Bwv

So $I = \frac{Bwv(Lh)\sigma}{w} = BLh\sigma v$

So $F_B = BIw = B^2Lh\sigma wv$

So net external force = $\Delta Phw - B^2Lh\sigma wv = kv$

(at steady state)

In the absence of magnetic field, $\Delta Phw = kv_0$

So $\Delta Phw - B^2 Lh\sigma wv = \frac{\Delta Phw}{v_0} v$

So $v = \frac{\Delta Pv_0}{\Delta P + B^2 L\sigma v_0}$

17. D

18. B

Sol. (for Q. 17-18)

Acceleration of ball, a = $\frac{qV}{md}$

So time of motion, $\Delta t = \sqrt{\frac{2d}{a}} = \sqrt{\frac{2md^2}{qV}}$

So $I = \frac{\Delta Q}{\Delta t} = \frac{n_0 qA}{\sqrt{\frac{2md^2}{qV}}} = \frac{C}{\varepsilon_0} n_0 q\sqrt{\frac{qV}{2m}}$

So $R = \frac{V}{I} = \frac{\varepsilon_0 V}{Cn_0 q\sqrt{\frac{2m}{qV}}} = \frac{\varepsilon_0}{Cn_0 q}\sqrt{\frac{2mV}{q}}$

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

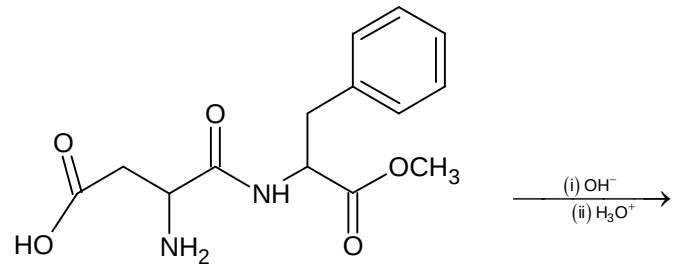
This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

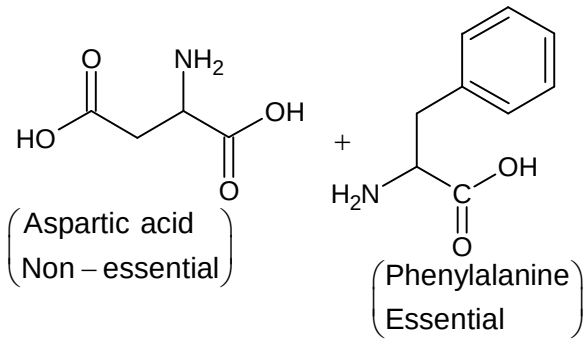
19. B

Sol. Down the group nucleophilicity generally increases in polar protic solvent, and if attacking atom is same stronger the base, stronger the nucleophile.

20. D

Sol.

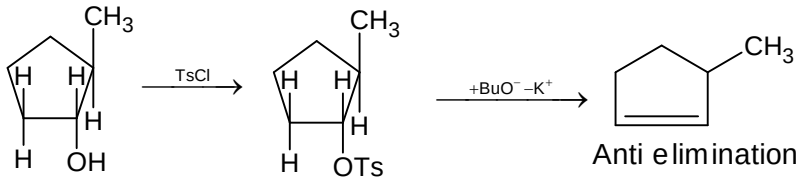




(Aspartic acid, Non – essential) (Phenylalanine, Essential)

21. B

Sol.



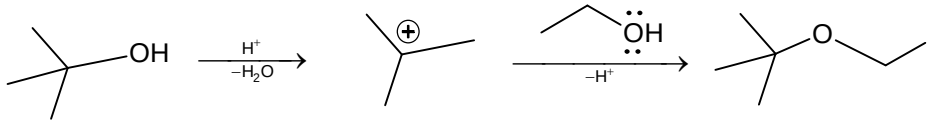
Anti elimination

22. A

Sol. The – COOH group is electron-withdrawing, acid strengthening compared to – H or – CH_3.

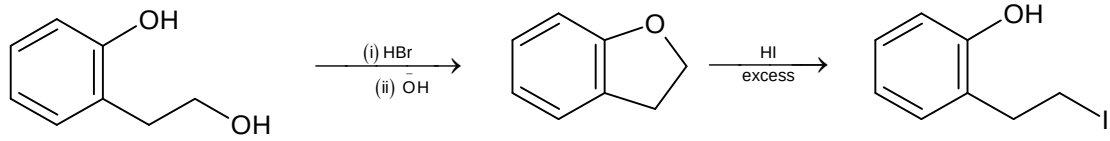
23. D

Sol.



24. A

Sol.



25. B

Sol. db is conjugated by two carbonyl groups.

26. D

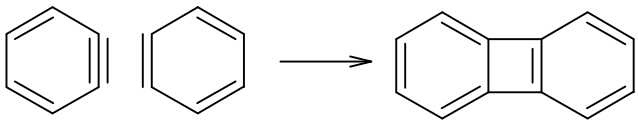
Sol. Conceptual.

27. A, B, C, D

Sol. Conceptual.

28. A, C, D

Sol.



Benzyne Biphenylene

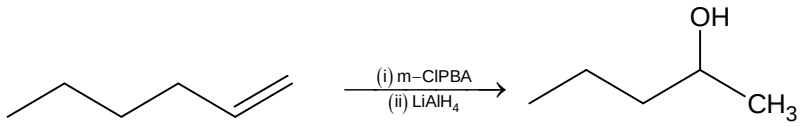
All C – C bonds are not equal in biphenylene. Central ring is antiaromatic.

29. B, D

Sol. Conceptual.

30. A, C

Sol.

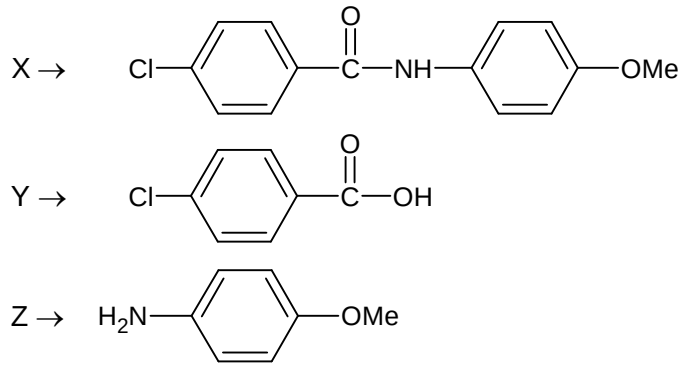


31. A, B, D

Sol. Conceptual.

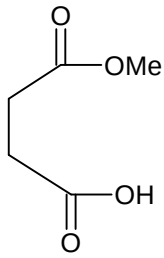
32. A, D

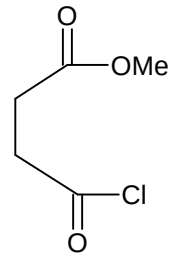
Sol.

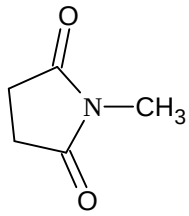


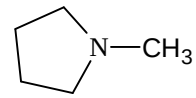
33. D

Sol.

(A) → 

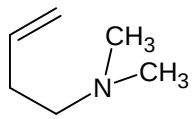
(B) → 

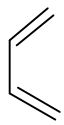
(C) → 

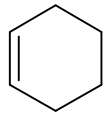
(D) → 

34. B

Sol.

(E) → 

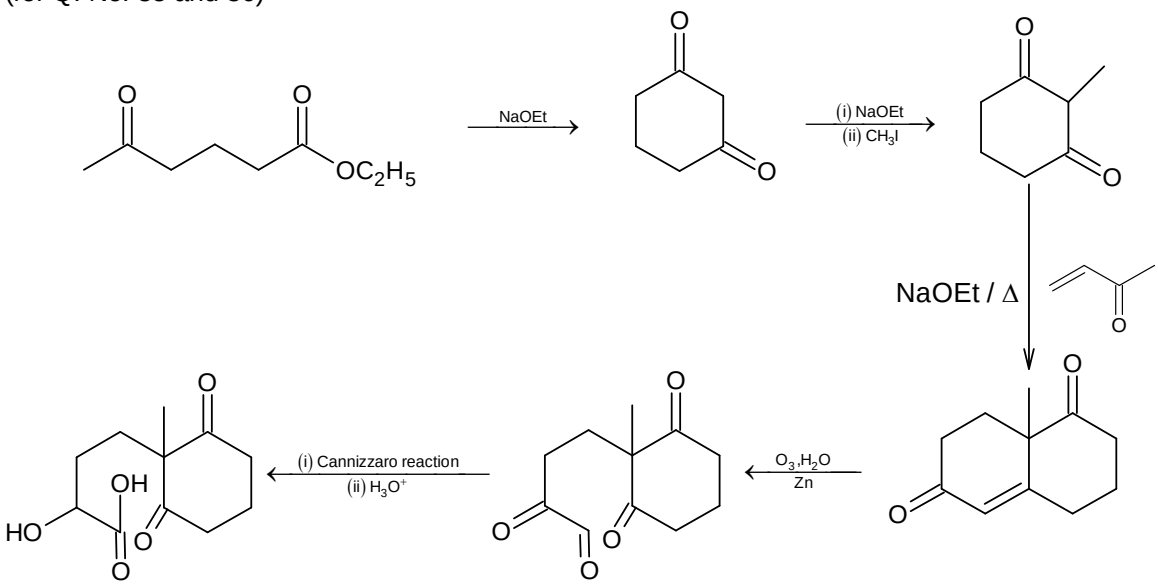
(F) → 

(H) → 

35. B

36. B

Sol. (for Q. No. 35 and 36)



Mathematics

PART – III

SECTION – A

37. C

Sol. $2020 \geq f(x) \geq 2\sqrt{2019}$

38. A

Sol. C always lies on circle

39. C

Sol. $\frac{1}{\sqrt{r_n}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_{n-1}}} \Rightarrow r_{2020} = \frac{r}{(2019)^2}$

40. C

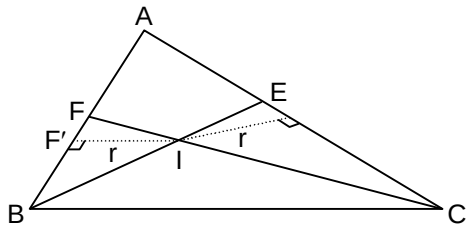
Sol. IF = $r \operatorname{cosec}\left(B + \frac{C}{2}\right)$

IE = $r \operatorname{cosec}\left(C + \frac{B}{2}\right)$

IE = IF $\Rightarrow \sin\left(B + \frac{C}{2}\right) = \sin\left(C + \frac{B}{2}\right)$

$\Rightarrow 2\sin\left(\frac{B}{4} - \frac{C}{4}\right)\cos\left(\frac{3B}{4} + \frac{3C}{4}\right) = 0$

$\Rightarrow \frac{3}{4}(B + C) = (2n+1)\frac{\pi}{2}$, $n \in I$



41. B

Sol. Let B_n be set of natural numbers less than or equal to 105 which are multiples of 'n', then

$A = B_1 - B_2 \cup B_5 \cup B_7$ ($\because$ $105 = 3 \times 5 \times 7$)

$\Rightarrow \sum_{r \in A} \cos\left(\frac{2\pi r}{105}\right) = \sum_{r=1}^{105} \cos\left(\frac{2\pi r}{105}\right) - \sum_{r \in B_3 \cup B_5 \cup B_7} \cos\left(\frac{2\pi r}{105}\right)$

Using sum of 'n' n^{th} roots of unity is 0, we get $\sum_{r \in A} \cos\left(\frac{2\pi r}{105}\right) = -1$

42. A

Sol. $\sum_{r=1}^{6} r^3 = 441$ if equation has a solution then sin(rπx) should be equal to 1 for r = 1, 2, ..., 6, which is not possible

43. A

Sol. Use $\cos 2\theta = 2\cos^2\theta - 1$

44. A, B

Sol. $g(x) = ax^2 + \frac{5}{4} - \left(\sin b - \frac{1}{2}\right)^2$

45. A, B, C

Sol. Let equation of asymptotes be $A : 2x^2 + 12xy - 7y^2 - 16x + 2y + d = 0$ where 'd' is given by $\Delta = 0$

Angle bisectors of asymptotes give principal axis

46. B, D

Sol. Length of chord joining $P(t_1)$ and $Q(t_2)$ is $|t_1 - t_2|\sqrt{(t_1+t_2)^2+4}$

As PQ is normal at P $\Rightarrow t_2 = \frac{-2}{t_1} - t_1$. Length is minimum for $t_1^2 = 2$ i.e., $t_1 = \pm\sqrt{2}$

$\Rightarrow$ P is $(2, 2\sqrt{2})$ or $(2, -2\sqrt{2})$, A is (–2, 0) and B is (4, 0)

Area of ΔPAB is $6\sqrt{2}$ sq. units and circumcenter is focus

47. A, B, C, D

Sol. For $\theta = \frac{\pi}{2}$, particle can pass through all point on coordinate axes

For $\theta = \frac{\pi}{6}$, if particle always changes its direction anticlockwise, then if will lie on $x - \sqrt{3}y - 8 = 0$ after 2019 seconds

48. A, D

Sol. $2^{\sin x - \cos x} = \pm\tan x \Rightarrow 2^{\sin x - \cos x} = \tan x \Rightarrow x = n\pi + \frac{\pi}{4}$, $n \in W$

49. A, C

Sol. Let tangent of slope m be $y = mx \pm \sqrt{m^2-1}$, then side length will be $2\sqrt{\frac{m^2-1}{m^2+1}}$

Also, length of diagonal of square should be greater than or equal to distance between points of tangency

$\Rightarrow 2\sqrt{2}\sqrt{\frac{m^2-1}{m^2+1}} \geq 2\sqrt{\frac{m^2+1}{m^2-1}} \Rightarrow m^2 \geq (\sqrt{2}+1)^2$

Side length = $2\sqrt{\frac{m^2-1}{m^2+1}}$ (for smallest square) = $2\sqrt{\frac{2+2\sqrt{2}}{4+2\sqrt{2}}} = 2^{\frac{3}{4}}$

Side length of largest possible square is 2 (as $m \to \infty$)

50. A, B, C

Sol. For distinct real lines $h^2 > ab$ and $g^2 > ac$
For coincident lines $h^2 = ab$ and $g^2 = ac$
For distinct parallel lines $h^2 > ab$, $g^2 > ac$

51. B

Sol. $A_1 = \pi$, $A_2 = 4\pi$, $A_3 = 9\pi$, $A_4 = 36\pi$

52. A

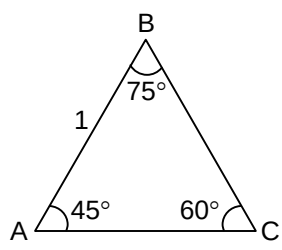
Sol. Radical centre is incenter of the triangle formed by three lines

53. C

Sol. Minimum perimeter is for orthic triangle whose sides are
a cos A, b cos B, c cos C

$a = \sqrt{\frac{2}{3}}$, b = 1, $c = \frac{\sqrt{3}+1}{\sqrt{6}}$ (Using sine rule)

So, sides of ΔDEF are $\frac{1}{2}$, $\frac{1}{\sqrt{3}}$ and $\frac{1}{2\sqrt{3}}$



54. B

Sol. Radius of circumcircle of ΔDEF = half of circumradius of ΔABC = $\frac{1}{2}\cdot\frac{1}{2\sin 60^\circ} = \frac{1}{2\sqrt{3}}$